

NE555 Test 4: Oscillating Output

Goal: Create an oscillating output with the 555 timer

Hypothesis: We can use a capacitor connected to pin 2 and 6 as a timer of sorts, we will follow a wiring schematic for this

Process:

1. Connect Vcc, GND, and Reset as normal
2. Connect a grounded 10nF capacitor to control
3. Create an electrical node with a grounded capacitor, trigger and threshold as well a resistor
4. Connect the resistor to a node with discharge and a resistor to Vcc
5. Connect output to an LED

Conclusion:

- When wired correctly it works perfect
- By changing capacitor size we can change the frequency of the blinking
 - this is because trigger and threshold measure based on divisions of V_{cc} so the larger the capacitor the longer it takes to fill and empty
- One thing I don't fully understand is what the resistors do
 - I'm going to do a test with those in the future
- Overall the test was a success